

# Development of a Deep Reinforcement Learning Agent for Optimal Project Portfolio Budgeting under Uncertain Cash Flows

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## Abstract

Project portfolio budget allocation for EPC contractors in the oil and gas sector is a sequential decision-making problem under uncertain cash inflows, cash outflows, and project performance. Each portfolio consists of multiple projects with heterogeneous attributes, including budget at completion, start and finish dates, contractual S-curves, fixed profit margins, milestone-based payment structures, and uncertain contractor performance represented through Earned Value Management metrics such as cumulative SPI and CPI. The model also captures delayed and uncertain milestone payments, working-capital shortages, credit usage during negative-cashflow periods, and management intervention actions that affect cashflow and performance trajectories.

Despite its practical importance, rigorous mathematical modeling of project portfolio budgeting under joint cashflow and performance uncertainty remains largely absent in recent literature, while most AI/ML-related studies discuss the topic qualitatively. This paper proposes a custom stochastic reinforcement learning environment for learning adaptive budget allocation policies under unknown future uncertainty. A PPO-based agent is trained with a multi-objective reward function to allocate constrained budgets efficiently, prioritize economically viable projects, avoid loss-making allocations, and learn optimal allocation timing. The framework provides a foundational RL model for EPC oil and gas portfolio budgeting, supports future fine-tuning with anonymized industry data, and enables immediate next-period allocation recommendations as well as rolling 12-period trajectory simulation.

**Keywords:** Project Portfolio Budgeting, Reinforcement Learning, EPC Contractors, Oil and Gas Projects, Cashflow Uncertainty, Earned Value Management, PPO, Rolling Horizon Control

## 1 Problem Formulation

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## 2 Problem Formulation

This section develops the formal problem structure for portfolio-level budget allocation in contracted project portfolios. The formulation proceeds across three levels. **Level 1** (Sec-

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tion ??) treats all contractor performance parameters as known and establishes the deterministic MILP whose optimum constitutes a theoretical upper bound: no implementable policy operating under uncertainty can exceed it. **Level 2** (Section ??) reintroduces stochastic efficiency and characterizes the resulting multi-stage stochastic program, establishing that its scenario tree grows exponentially in the number of projects and periods. **Level 3** (Section ??) reformulates the intractable stochastic program as a Constrained Markov Decision Process (CMDP), providing an explicit mapping from LP elements to MDP primitives and yielding a Gymnasium-compatible environment on which the PPO agent of Section ?? is trained. The three levels are a coherent arc — Level 1 defines the ceiling, Level 2 explains why it is unreachable, Level 3 provides the tractable path — all sharing the primitives defined next.

## 2.1 Scope, Assumptions, and Problem Primitives

**Scope.** The problem is operational, not strategic: the portfolio of contracted projects is fixed, commitments are signed, and the decision-maker exercises a single lever — the budget  $x_{i,t}$  directed to each active project  $i$  at each discrete period  $t$ . All outcomes of interest (milestone achievement, cash balance, schedule adherence, termination) flow through this allocation decision. Portfolio selection is treated as exogenous ?.

**Structural assumptions.** Four choices underlie the entire formulation. (i) *Independent bilateral contracts*: projects interact only through the shared budget constraint; no resource or workforce sharing occurs, consistent with standard EPC contracting practice. (ii) *Budget as the sole control*: the contractor’s internal resource decisions are absorbed into the efficiency parameter  $\eta_{i,t}$ ; the portfolio manager observes only how much physical progress a budget allocation yields, not how it is achieved. (iii) *Discrete monthly periods*:  $\mathcal{T} = \{1, \dots, H\}$  reflects the monthly governance cycle of portfolio management (progress reporting, payment certification, performance review). (iv) *FIDIC contract mechanics*: the milestone payment structure, advance payment and recovery, retention, and termination settlement are parameterized consistently with FIDIC Red, Yellow, and Silver Book forms ?; all contract-specific quantities are calibrated parameters, not fixed constants.

### 2.1.1 Sets, Indices, and the Decision Variable

[leftmargin=2.2cm, labelwidth=2.0cm]Projects  $\{1, \dots, n\}$ , indexed by  $i$ ; all committed at horizon start. Planning periods  $\{1, \dots, H\}$ , indexed by  $t$  (calendar months). Ordered milestones  $\{1, \dots, M_i\}$  for project  $i$ , indexed by  $j$ , with thresholds  $\theta_{i,1} < \dots < \theta_{i,M_i} = 1$ . Planned execution window  $\{s_i, \dots, f_i\}$ ; anchors S-curve computations. Projects active at period  $t$ ; fixed parameter at Level 1, endogenous at Levels 2–3. Milestones of project  $i$  certified by period  $t$ :  $\{j : \theta_{i,j} \leq P_{i,t}^{\text{actual}}\}$ . **Decision variable**: budget allocated to project  $i$  at period  $t$ . The period- $t$  action is  $\mathbf{x}_t = (x_{i,t})_{i \in \mathcal{P}_t^{\text{active}}}$ . Portfolio cash balance at the start of period  $t$ ;  $B_1$  is given.

### 2.1.2 Project Parameters

Tables ?? and ?? list all project parameters, fixed at commitment. The planned cumulative expenditure fraction follows a normalized S-curve ?? parameterized by  $(a_i, b_i)$ :

$$P_{i,t}^{\text{plan}} = F_i(\tau_{i,t}^c) = I_{\tau_{i,t}^c}(a_i, b_i), \quad \tau_{i,t} = \frac{t - s_i + 1}{D_i^{\text{plan}}}, \quad \tau_{i,t}^c = \min\{1, \max\{0, \tau_{i,t}\}\},$$

where  $I_\tau(\cdot, \cdot)$  is the regularized incomplete Beta function. The planned period expenditure increment is  $\Delta \bar{E}_{i,t} = \text{BAC}_i [F_i(\tau_{i,t}^c) - F_i(\tau_{i,t-1}^c)]$ . This profile is a fixed parameter derived at commitment from  $(a_i, b_i, s_i, f_i, \text{BAC}_i)$ ; it is never a decision variable.

| Parameter                                     | Symbol              | Domain           | Definition                                                                                         |
|-----------------------------------------------|---------------------|------------------|----------------------------------------------------------------------------------------------------|
| <b>Planned start &amp; finish</b><br>Schedule | $s_i, f_i$          | $\mathbb{Z}^+$   | Anchor of execution window $\mathcal{A}_i^{\text{plan}}$ and S-curve.                              |
| <b>Planned duration</b><br>Schedule           | $D_i^{\text{plan}}$ | $\mathbb{Z}^+$   | $f_i - s_i + 1$ ; normalizes progress against plan.                                                |
| <b>S-curve shape</b><br>Schedule              | $a_i, b_i$          | $\mathbb{R}^+$   | Beta CDF parameters governing the planned expenditure profile.                                     |
| <b>Milestone threshold</b><br>Schedule        | $\theta_{i,j}$      | $(0, 1]$         | Cumulative progress fraction required to certify milestone $j$ .                                   |
| <b>Delay buffer</b><br>Schedule               | $\text{buffer}_i$   | $\mathbb{Z}_0^+$ | Periods beyond $f_i$ before forced termination is assessed.                                        |
| <b>Contract price</b><br>Contract             | $CP_i$              | $\mathbb{R}^+$   | Total client payment; $CP_i = \text{BAC}_i(1 + \pi_i)$ .                                           |
| <b>Profit margin</b><br>Contract              | $\pi_i$             | $\mathbb{R}_0^+$ | Contractual markup on $\text{BAC}_i$ .                                                             |
| <b>Advance payment ratio</b><br>Contract      | $\alpha_i$          | $[0, 1]$         | $A_i = \alpha_i \cdot CP_i$ disbursed at $t = s_i$ ; recovered via milestone deductions.           |
| <b>Retention ratio</b><br>Contract            | $\rho_i$            | $[0, 1]$         | Fraction withheld per milestone; released as $R_i^{\text{ret}} = \rho_i \cdot CP_i$ at completion. |
| <b>Milestone weight</b><br>Contract           | $\phi_{i,j}$        | $\mathbb{R}^+$   | Fraction of $CP_i$ at milestone $j$ ; $\sum_j \phi_{i,j} = 1$ .                                    |

Table 1: Schedule and contract parameters for project  $i \in \mathcal{P}$  (fixed at commitment).

| Parameter                                   | Symbol      | Domain           | Definition                                                                                                                                                                       |
|---------------------------------------------|-------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Budget at completion</b><br>Cost         | $BAC_i$     | $\mathbb{R}^+$   | Total planned contractor expenditure; EVM cost basis.                                                                                                                            |
| <b>Discount factor</b><br>Cost              | $\gamma$    | $(0, 1]$         | Per-period discount applied to future cash flows; shared across $\mathcal{P}$ .                                                                                                  |
| <b>Initial cash balance</b><br>Cost         | $B_1$       | $\mathbb{R}^+$   | Portfolio cash balance at horizon start; initial condition of balance recursion.                                                                                                 |
| <b>Deviation penalty weight</b><br>Penalty  | $\lambda_i$ | $\mathbb{R}_0^+$ | Per-period cost per unit absolute progress deviation; calibrated so cumulative penalty at planned finish under sustained contractual-limit delay equals $\Lambda_i \cdot CP_i$ . |
| <b>Deviation penalty cap</b><br>Penalty     | $\Lambda_i$ | $\mathbb{R}_0^+$ | Maximum cumulative penalty as fraction of $CP_i$ ; applies symmetrically to lag and acceleration.                                                                                |
| <b>Termination threshold</b><br>Termination | $\kappa_i$  | $(0, 1]$         | Fraction of $\Lambda_i \cdot CP_i$ at which project enters termination candidacy.                                                                                                |
| <b>Minimum allocation</b><br>Termination    | $m_i^0$     | $\mathbb{R}_0^+$ | Floor budget to hold project in contractually serviceable state while termination conditions are evaluated; no performance-recovery role.                                        |
| <b>Confirmation window</b><br>Termination   | $k$         | $\mathbb{Z}^+$   | Consecutive periods at floor $m_i^0$ required to confirm abandonment.                                                                                                            |

Table 2: Cost, penalty, and termination parameters for project  $i \in \mathcal{P}$  (fixed at commitment).

### 2.1.3 State Variables and Transition Dynamics

**Project-level state.** For each  $i \in \mathcal{P}$ ,  $t \in \mathcal{T}$ :  $P_{i,t}^{\text{actual}} \in [0, 1]$  (cumulative progress),  $E_{i,t} \geq 0$  (cumulative expenditure),  $SPI_{i,t} = P_{i,t}^{\text{actual}}/P_{i,t}^{\text{plan}}$ ,  $CPI_{i,t} = P_{i,t}^{\text{actual}} \cdot BAC_i/E_{i,t}$ ,  $\delta_{i,t} = P_{i,t}^{\text{actual}} - P_{i,t}^{\text{plan}}$  (signed deviation),  $\Delta_{i,t}^{\text{pen}} \geq 0$  (cumulative penalty), and  $\text{status}_{i,t} \in \{\text{active}, \text{completed}, \text{terminated}\}$ . The portfolio state at period  $t$  is:

$$S_t = \left( \{P_{i,t}^{\text{actual}}, E_{i,t}, SPI_{i,t}, CPI_{i,t}, \delta_{i,t}, \Delta_{i,t}^{\text{pen}}, \text{status}_{i,t}\}_{i \in \mathcal{P}}, B_t \right).$$

**Budget-to-progress conversion.** Budget  $x_{i,t}$  converts to progress through the efficiency  $\eta_{i,t}$ :

$$P_{i,t}^{\text{actual}} = P_{i,t-1}^{\text{actual}} + \eta_{i,t} \cdot \frac{x_{i,t}}{BAC_i}, \quad E_{i,t} = E_{i,t-1} + x_{i,t}, \quad (1)$$

with  $P_{i,0}^{\text{actual}} = E_{i,0} = 0$ . The scalar  $\eta_{i,t}$  absorbs all sources of conversion uncertainty (productivity, supply chain, weather). At Level 1,  $\eta_{i,t}$  is replaced by the known deterministic value  $\bar{\eta}_{i,t}$  (equal to unity under nominal plan); at Levels 2–3 it is a stochastic primitive realized only after  $x_{i,t}$  is committed.

**Contractual cash-flow mechanics.** Payment inflows follow FIDIC mechanics ?. At initiation ( $t = s_i$ ), the advance payment  $A_i = \alpha_i \cdot CP_i$  is received. Milestone  $j$  is certified

at the first  $t$  with  $P_{i,t}^{\text{actual}} \geq \theta_{i,j}$ ; the net payment (after retention and advance recovery) is:

$$R_{i,j}^{\text{net}} = \phi_{i,j} \cdot CP_i(1 - \rho_i) - \phi_{i,j} \cdot A_i. \quad (2)$$

Upon completion ( $P_{i,t}^{\text{actual}} = 1$ ) the accumulated retention  $R_i^{\text{ret}} = \rho_i \cdot CP_i$  is released together with the final milestone payment. The payment identity  $A_i + \sum_j R_{i,j}^{\text{net}} + R_i^{\text{ret}} = CP_i$  holds exactly. Aggregate period inflow is:

$$V_t = \sum_{i,j} R_{i,j}^{\text{net}} \cdot \mathbf{1}\{(i,j) \text{ newly certified at } t\} + \sum_{i \in \mathcal{P}_t^{\text{term}}} R_i^{\text{term}}, \quad (3)$$

where  $\mathcal{P}_t^{\text{term}}$  collects projects newly terminated at  $t$ , with settlement  $R_i^{\text{term}}$  defined in (??) below.

**Symmetric running deviation penalty.** Schedule adherence is priced at every period:

$$c_{i,t}^{\text{dev}} = \min\{\lambda_i |\delta_{i,t}|, \Lambda_i \cdot CP_i\}, \quad \Delta_{i,t}^{\text{pen}} = \sum_{\tau=s_i}^t c_{i,\tau}^{\text{dev}}. \quad (4)$$

The penalty is symmetric ( $|\delta_{i,t}|$ , not  $\max\{0, -\delta_{i,t}\}$ ): both lag and acceleration are penalized, discouraging budget-pushing that risks quality degradation ?. The cap  $\Lambda_i \cdot CP_i$  mirrors the contractual delay-penalty ceiling under FIDIC terms.

**Termination conditions and settlement.** A project transitions to **terminated** at the first period  $t$  at which both conditions hold simultaneously:

$$(i) \text{ Penalty saturation: } \Delta_{i,t}^{\text{pen}} \geq \kappa_i \cdot \Lambda_i \cdot CP_i, \quad (5)$$

$$(ii) \text{ Confirmed floor: } x_{i,\tilde{t}} \leq m_i^0 \quad \forall \tilde{t} \in \{t - k + 1, \dots, t\}. \quad (6)$$

Condition (??) signals exhaustion of contractual schedule tolerance; condition (??) confirms that the portfolio manager has deliberately ceased investment rather than transiently reduced it. Upon termination at  $t_i^{\text{term}}$ , the net settlement is:

$$R_i^{\text{term}} = P_{i,t_i^{\text{term}}}^{\text{actual}} \cdot CP_i(1 - \rho_i) - \left( A_i - \sum_{j \in \mathcal{M}_i^{\text{cert}}(t_i^{\text{term}})} \phi_{i,j} \cdot A_i \right). \quad (7)$$

Negative  $R_i^{\text{term}}$  (outstanding advance exceeds earned value) constitutes a net obligation recovered from portfolio inflows, consistent with FIDIC termination clauses.

**Portfolio cash balance recursion.**

$$B_{t+1} = B_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} + V_t, \quad B_t \geq 0 \quad \forall t \in \mathcal{T}. \quad (8)$$

The constraint  $B_t \geq 0$  prohibits borrowing; the portfolio operates under a fixed capital commitment ?. The shared balance  $B_t$  is the sole coupling across otherwise independent project dynamics, and its linear scaling with  $n$  characterizes the complexity that motivates the intractability argument in Level 2.

### 2.1.4 Shared Objective

The decision-maker maximizes discounted net cash flow over the horizon:

$$\max_{\{x_{i,t}\}} \mathbb{E} \left[ \sum_{t=1}^H \gamma^{t-1} \left( V_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} - \sum_{i \in \mathcal{P}_t^{\text{active}}} c_{i,t}^{\text{dev}} \right) \right], \quad (9)$$

subject to, for all  $i \in \mathcal{P}$ ,  $t \in \mathcal{T}$ :

$$x_{i,t} \geq 0, \quad (10)$$

$$\sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} \leq B_t, \quad (11)$$

$$x_{i,t} \geq m_i^0 \cdot \mathbf{1}\{\text{status}_{i,t} = \text{active}\}. \quad (12)$$

There is no terminal sum. Completion incentives are embedded structurally:  $R_i^{\text{ret}}$  and  $R_{i,M_i}^{\text{net}}$  enter  $V_t$  at the certification period;  $R_i^{\text{term}}$  (possibly negative) enters at termination, creating a natural deterrent against premature abandonment. Constraint (??) ensures each active project remains in a contractually serviceable state; it is vacuous once  $\text{status}_{i,t} \neq \text{active}$ . The expectation  $\mathbb{E}[\cdot]$  collapses to a deterministic evaluation at Level 1 and is retained at Levels 2 and 3.

## 2.2 Level 1: Deterministic Full-Foresight MILP

Setting  $\eta_{i,t} = \bar{\eta}_{i,t}$  (known) renders  $S_t$  a deterministic function of  $\mathbf{x}_t$ , so  $\mathcal{P}_t^{\text{active}}$ , milestone certification times, and termination periods  $t_i^{\text{term}}$  are all computable ex ante. Introducing the certification binary  $u_{i,j,t} \in \{0,1\}$  ( $= 1$  iff milestone  $j$  of project  $i$  certified by period  $t$ ) and activity indicator  $\text{act}_{i,t} \in \{0,1\}$  (fixed parameter), the stochastic objective (??) collapses to the MILP:

$$\max \sum_{t=1}^H \gamma^{t-1} \left( V_t^{\text{det}} - \sum_{i \in \mathcal{P}} \text{act}_{i,t} x_{i,t} - \sum_{i \in \mathcal{P}} \text{act}_{i,t} c_{i,t}^{\text{dev}} \right) \quad (13a)$$

$$\text{s.t. } B_{t+1} = B_t - \sum_i \text{act}_{i,t} x_{i,t} + V_t^{\text{det}}, \quad B_t \geq 0, \quad \forall t \quad (13b)$$

$$\sum_i \text{act}_{i,t} x_{i,t} \leq B_t, \quad \forall t \quad (13c)$$

$$P_{i,t}^{\text{actual}} = P_{i,t-1}^{\text{actual}} + \bar{\eta}_{i,t} \frac{x_{i,t}}{\text{BAC}_i}, \quad P_{i,t}^{\text{actual}} \leq 1, \quad \forall i, t \quad (13d)$$

$$P_{i,t}^{\text{actual}} \geq \theta_{i,j} u_{i,j,t} - \mathbf{M}(1 - u_{i,j,t}), \quad \forall i, j, t \quad (13e)$$

$$P_{i,t}^{\text{actual}} \leq \theta_{i,j} + \mathbf{M} u_{i,j,t}, \quad u_{i,j,t} \geq u_{i,j,t-1}, \quad u_{i,j+1,t} \leq u_{i,j,t}, \quad \forall i, j, t \quad (13f)$$

$$\begin{aligned} V_t^{\text{det}} = & \sum_{i,j} R_{i,j}^{\text{net}} (u_{i,j,t} - u_{i,j,t-1}) + \sum_i R_i^{\text{ret}} (u_{i,M_i,t} - u_{i,M_i,t-1}) \\ & + \sum_i A_i \cdot \mathbf{1}\{t=s_i\} + \sum_{i \in \mathcal{P}^{\text{term}}} R_i^{\text{term}} \cdot \mathbf{1}\{t=t_i^{\text{term}}\}, \quad \forall t \end{aligned} \quad (13g)$$

$$\delta_{i,t} = P_{i,t}^{\text{actual}} - P_{i,t}^{\text{plan}} = d_{i,t}^+ - d_{i,t}^-, \quad d_{i,t}^+, d_{i,t}^- \geq 0, \quad \forall i, t \quad (13h)$$

$$c_{i,t}^{\text{dev}} \geq \lambda_i (d_{i,t}^+ + d_{i,t}^-), \quad c_{i,t}^{\text{dev}} \leq \Lambda_i \cdot CP_i, \quad c_{i,t}^{\text{dev}} \geq 0, \quad \forall i, t \quad (13i)$$

$$x_{i,t} \geq m_i^0 \cdot \text{act}_{i,t}, \quad \forall i, t \quad (13j)$$

$$x_{i,t} \geq 0, \quad u_{i,j,t} \in \{0,1\}, \quad u_{i,j,0} = P_{i,0}^{\text{actual}} = 0. \quad (13k)$$

The big- $M$  constant in (??)–(??) is  $\mathbf{M} = \max_{i,j} \theta_{i,j}$ ; tighter values may be derived project-specifically. Constraints (??)–(??) implement the exact if-and-only-if certification logic with monotone, ordered binary activation. Constraints (??)–(??) linearize the absolute-value and cap in (??): the maximizing objective drives  $c_{i,t}^{\text{dev}}$  to the tighter of the two bounds, recovering the exact symmetric penalty. The difference  $(u_{i,j,t} - u_{i,j,t-1})$  in (??) equals one in the sole period a milestone is first certified, linearizing the trigger without additional binaries. Termination timing  $t_i^{\text{term}}$  and the settlement  $R_i^{\text{term}}$  (??) are pre-computed from  $\bar{\eta}_{i,t}$  before solving; the termination conditions (??)–(??) are parameter-verifiable at Level 1 and become binding optimization constraints at Levels 2–3.

**Upper bound.** Let  $Z_{\text{L1}}^*$  denote the Level 1 optimum. For any causal policy  $\pi$  feasible under uncertainty,  $\sup_{\pi} \mathbb{E}[Z^{\pi}] \leq Z_{\text{L1}}^*$ : every feasible causal allocation sequence is also feasible in the Level 1 program solved on the realized trajectory, so the full-foresight optimum dominates in expectation. The gap  $Z_{\text{L1}}^* - Z^{\pi}$  quantifies the cost of uncertainty and anchors the empirical comparison in Section ??.